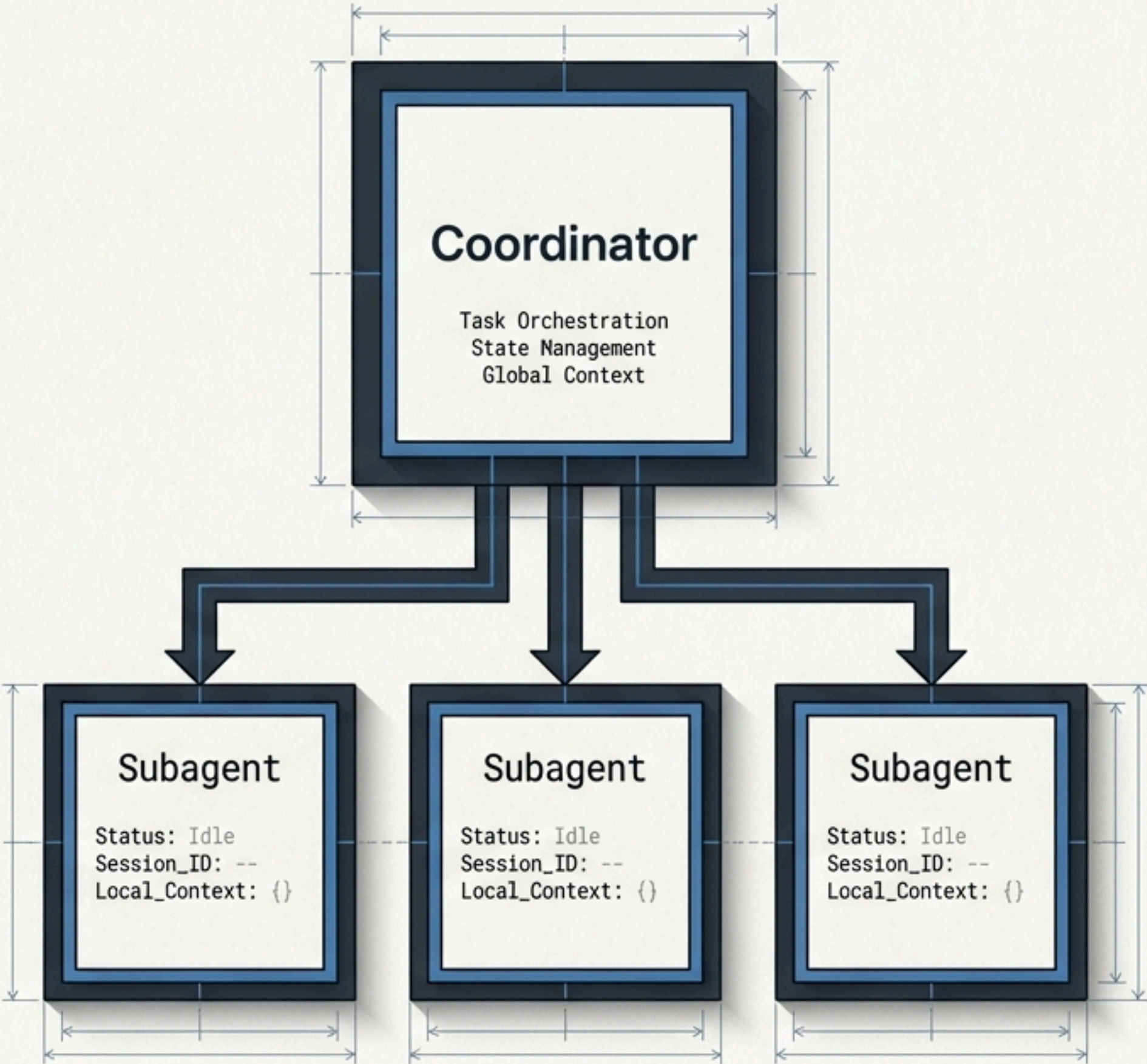


Agentic Architecture: Subagent Context & Session Management

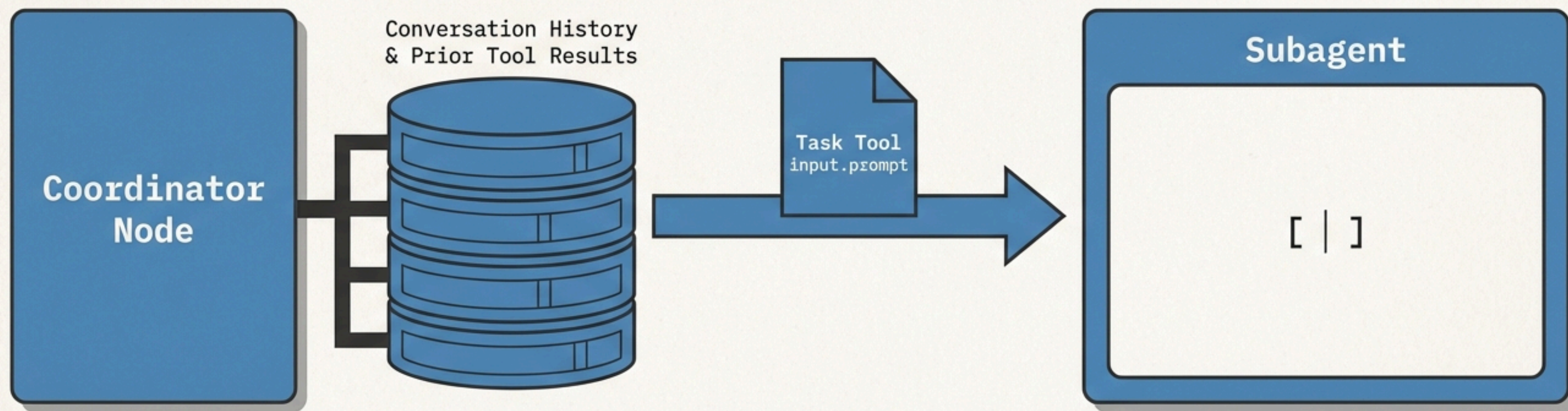
Claude Certified Architect | Module 1 · Domain 1

Episode:	03 of 20
Target_Audience:	Intermediate Developers & Architects
Exam_Weight:	Domain 1 (27% of exam)
Target_Tasks:	1.3 (Configure invocation), 1.7 (Manage session state)
Relevant_Scenarios	Scenario 2 (Code Gen), Scenario 3 (Multi-Agent Research)



Your Subagent Wakes Up Blank

The Amnesia Diagram



WARNING: Subagents DO NOT inherit the coordinator's history, prior tool results, previously discovered findings, or any intermediate state.

The Failure Pattern: Sending a vague instruction like "synthesize the previous findings" guarantees hallucinated or generic output because the subagent has zero access to those findings. Every subagent spawned via the Task tool starts with an entirely empty context.

Why Prose Handoffs Fail

Attribution Loss:

Source attribution is silently dropped during summarization.

10 Web Search Sources

→ Summarized into a paragraph

→ Passed to Synthesis Agent

→ Final Report written

→ Zero sources remain traceable.



Architectural Rule:

Summarization naturally destroys metadata unless the output schema explicitly forbids it.

Mixing findings as plain prose destroys the ability to route or verify them programmatically downstream.

Rule #1: Separate Content from Metadata

Never pass findings as plain prose. Always use typed structured objects.

Anatomy of a Finding Object

Structured Object
Context Boundary

Content Block
(For reasoning & synthesis)

`finding (string): The actual discovered content.`

Metadata Block
(For citation, routing, temporal validation)

`source_url (string): Originating document URL.`

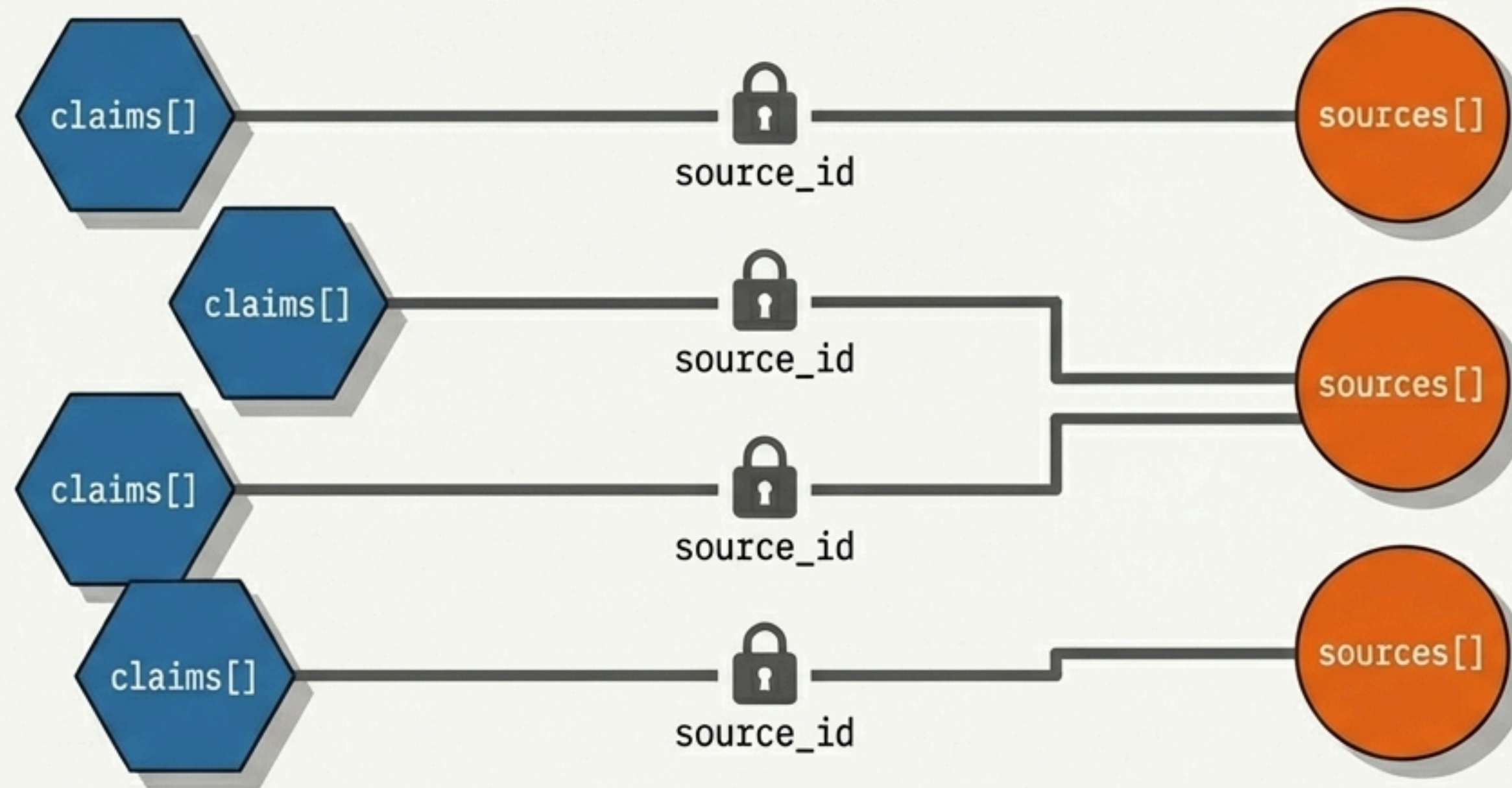
`source_title (string): Human-readable title.`

`page_number (int or null): Specific page reference.`

`retrieved_at (ISO 8601 string): Critical for temporal validation.`

`confidence (string): Subagent's internal rating.`

Preserving Provenance: Claim-Source Mappings

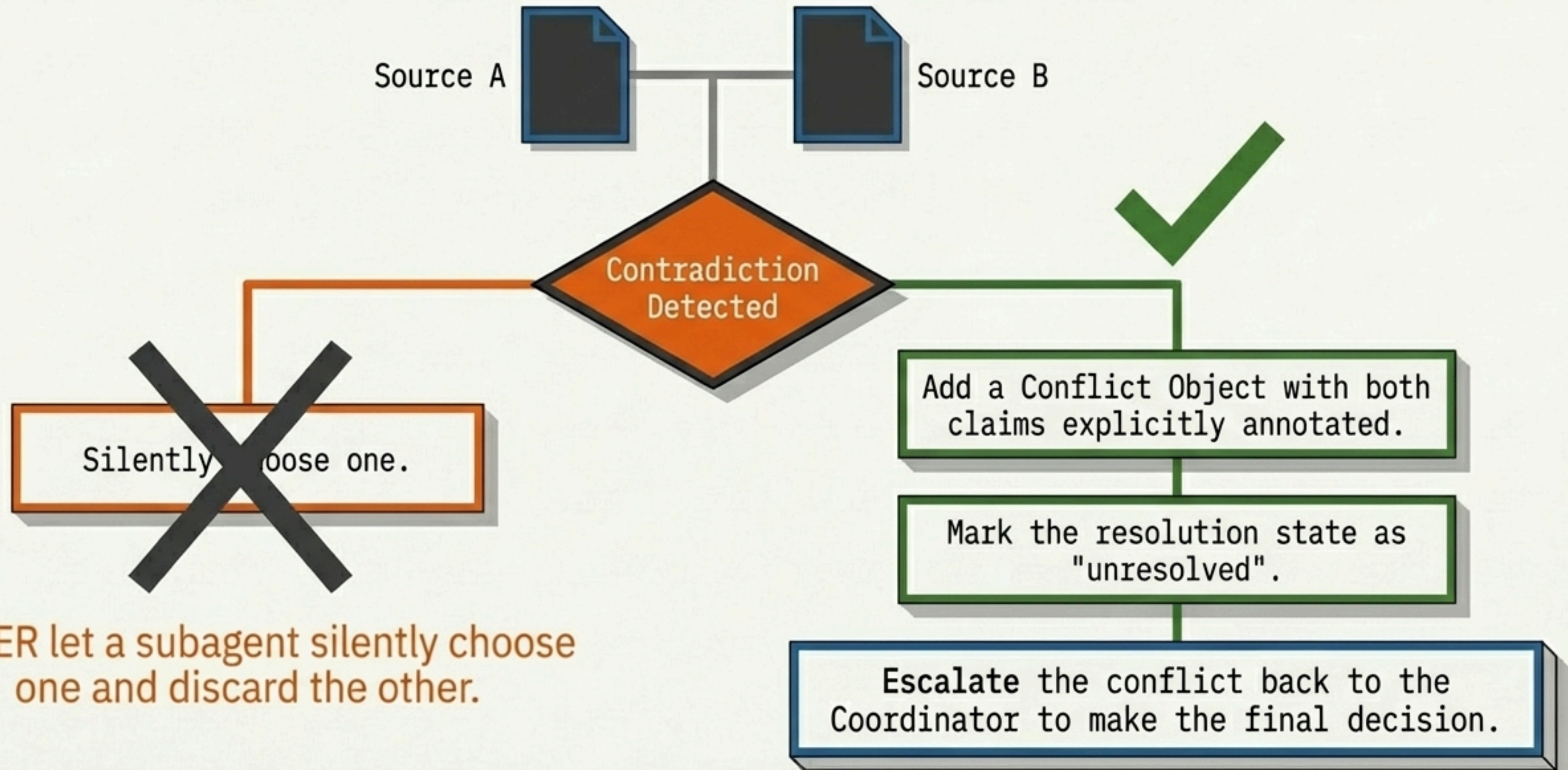


Core Mechanism: Every finding is stored as a specific claim object permanently linked to a source object via a shared ID.

Exam-Critical Rule: Every synthesis step in the pipeline MUST output both arrays intact. If the synthesis agent consumes structured data but outputs prose, the provenance chain is broken.

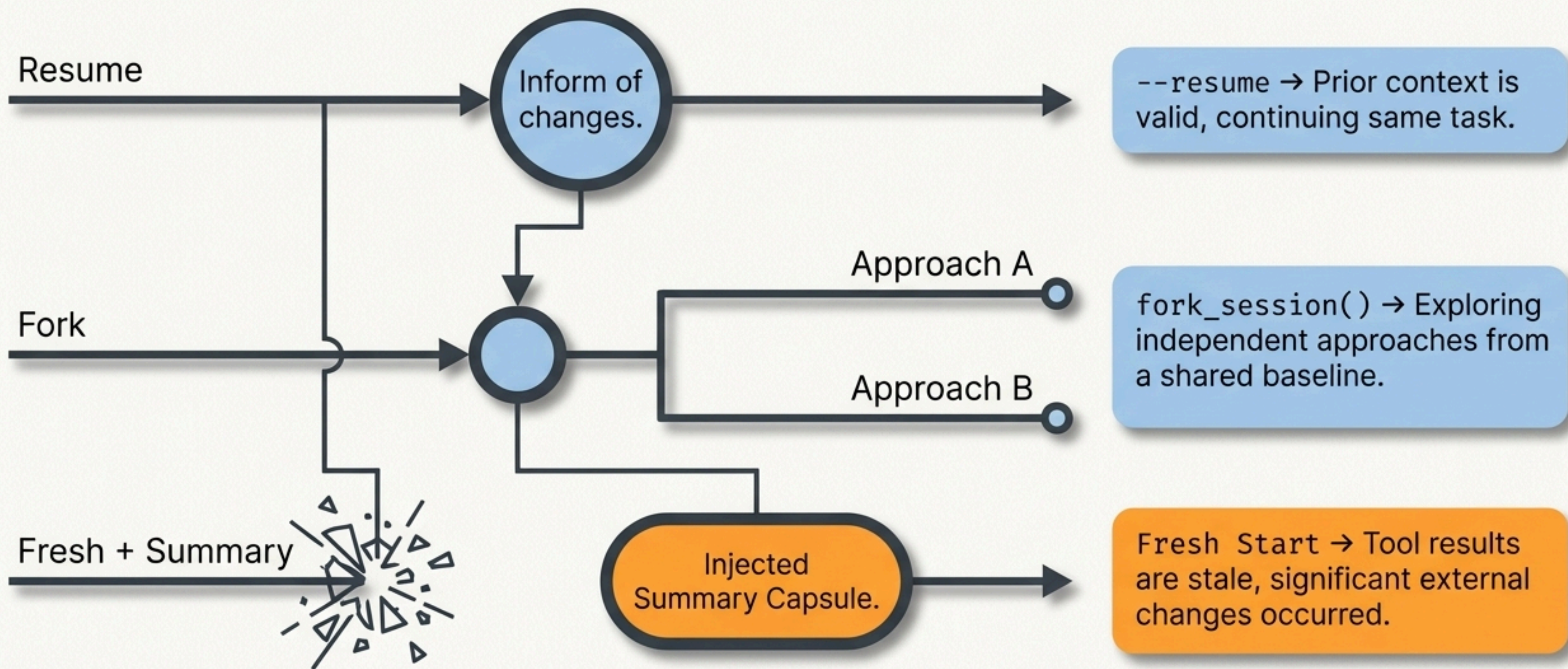
Resolving Reality Clashes

The Scenario: Two parsed sources say different things.



Managing State Across Time (Task 1.7)

Introduction to the **three core patterns** of session continuity.



Pattern 1: Named Session Resumption

CLI Command:

```
--resume <session-name>
```

Prompt

The Trap: Assuming the agent knows what you did while it was asleep. Agents cannot detect external file changes automatically.

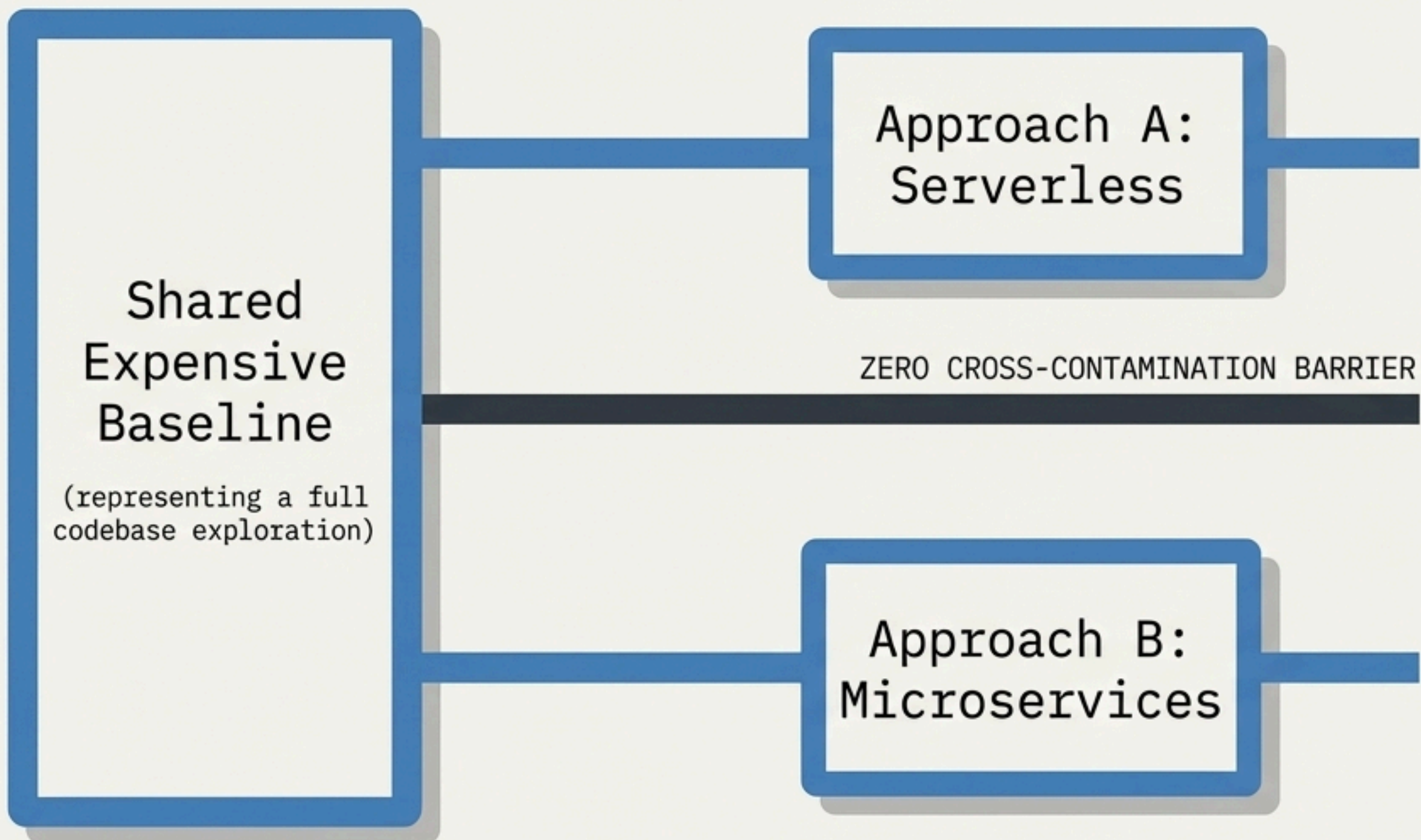


The Mandate: Always explicitly inform a resumed session about exactly what changed since the last run. If you resume without informing, the agent makes decisions based on an **outdated reality (stale tool results)**.

Since our last session, I have refactored the payments module. Do not rely on prior tool results for `auth.py` or `stripe_handler.py`.

Pattern 2: Fork Session

Function: **fork_session**(<session-id>)

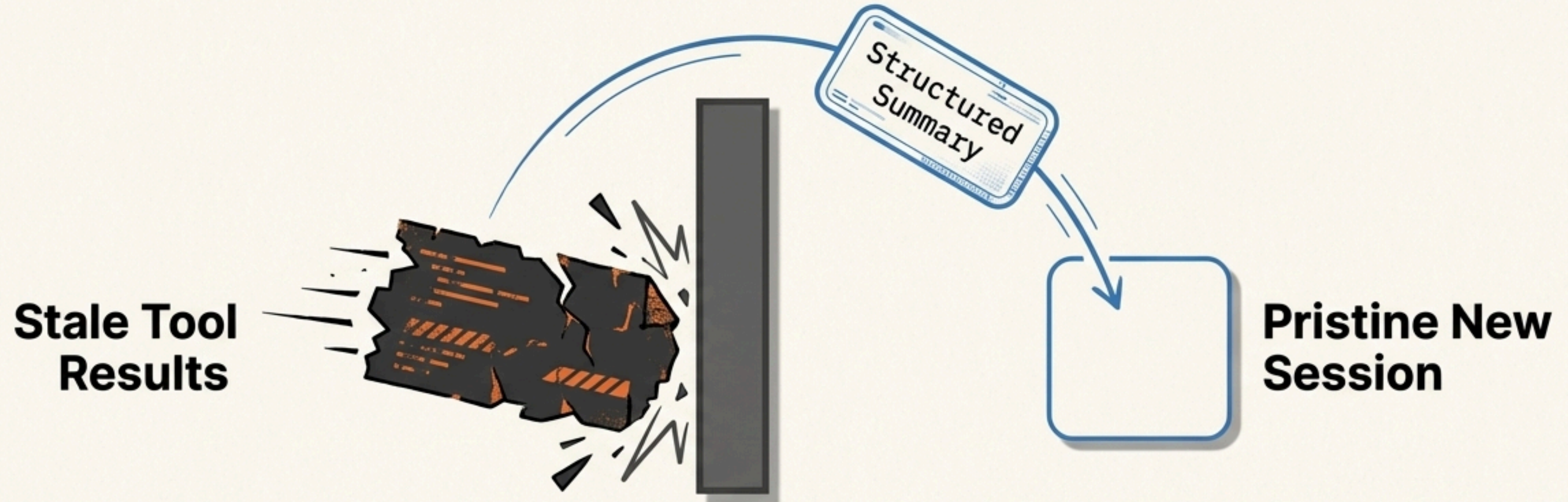


The Mechanism: Creates an independent branch from an existing session. Both branches share the expensive upfront work, but diverge independently.

Primary Use Case: Comparing two architectural approaches on the exact same codebase baseline without running full exploration twice.

! Exam Warning: NEVER use `fork_session` to handle targeted file updates over time.

Pattern 3: Fresh Start + Injection



The Trigger:

When tool results are fundamentally stale, resuming is actively counterproductive. Start fresh, but compress prior findings into an injected summary in the initial prompt.

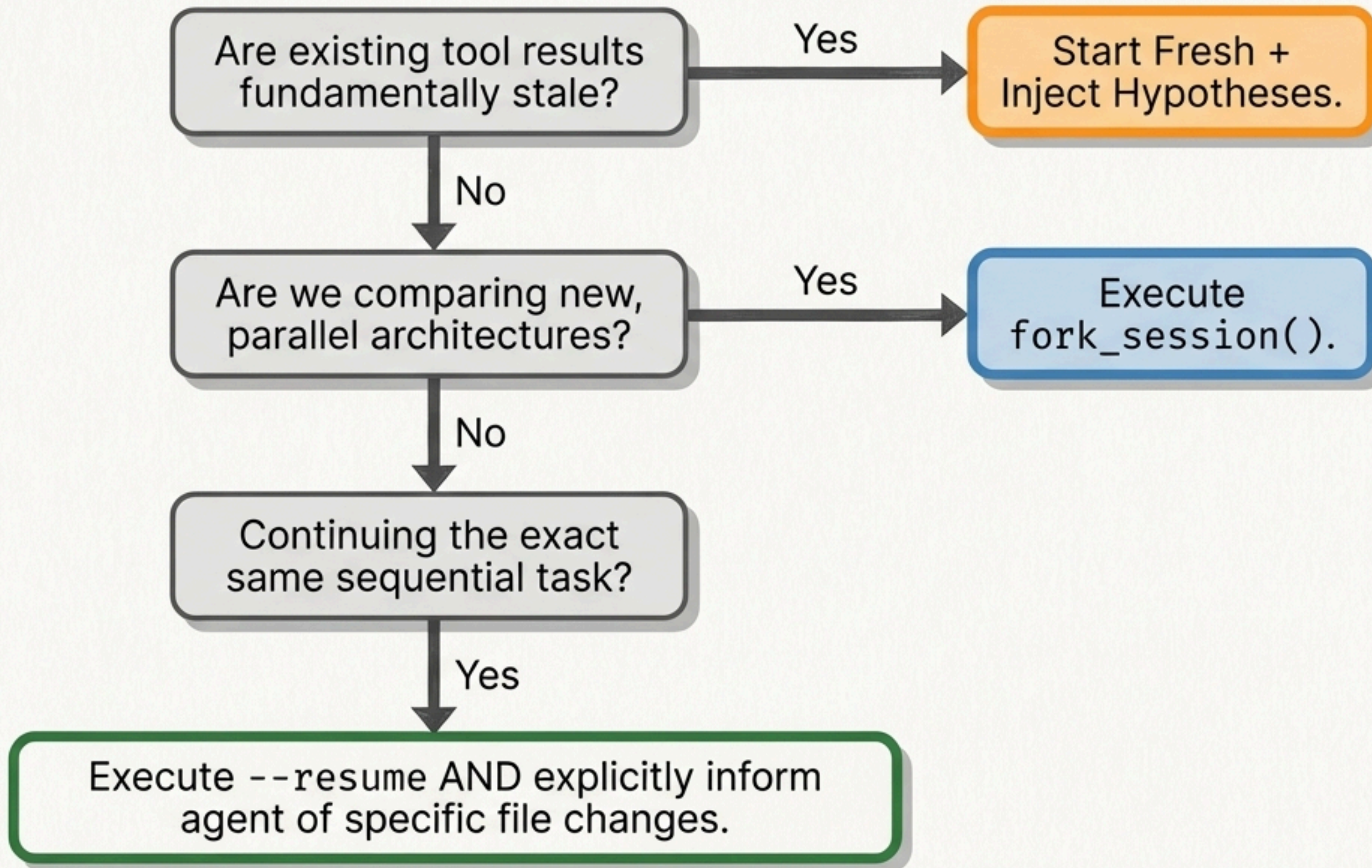
Critical Framing Rule:

The prompt must frame the injected summary as 'hypotheses to validate, not established facts.' This psychological cue prevents the model from blindly accepting information that may have changed.

The Session Routing Diagnostic

Current Context State	Codebase Volatility	Required Action Command
Context is Valid / Shared	Zero / Low Volatility	Use <code>--resume</code> + explicitly state updates.
Need Parallel Exploration	Shared Baseline context	Use <code>fork_session()</code> .
Tool Results are Stale	High Volatility / Time Passed	Use Fresh Session + Inject structured hypotheses.

Session Management Action Flowchart



Architectural Anti-Patterns (Exam Constraints)



NOT THAT: Passing **prose summaries** between agents (silently drops attribution).



DO THIS: Pass **structured context** via **Finding Objects**.



NOT THAT: Letting a subagent **silently resolve source conflicts** (destroys provenance).



DO THIS: **Annotate** both, mark **"unresolved,"** escalate to **Coordinator**.



NOT THAT: Using **fork_session** to handle **targeted file updates**.



DO THIS: **Fork** is for parallel exploration only.



NOT THAT: Resuming a session with a **heavily changed** codebase.



DO THIS: **Start fresh**, inject **hypotheses**.



NOT THAT: Resuming without **informing** the agent of changes.



DO THIS: Explicitly **declare** updated reality.

Sample Scenario Breakdown

Scenario 3 (Multi-Agent Research)

The Bug:

Claim attribution is lost because web results are summarized into prose before synthesis.

The Fix:

Require `structured objects` with explicit `claim_id` → `source_id` mappings. Require the `synthesis output schema` to preserve them.

Scenario 2 (Code Generation)

The Bug:

Codebase analysis ran 3 days ago. Payments module was just refactored.

The Fix:

Resume via `--resume`. Explicitly inform the agent which `specific files changed` to trigger `targeted re-analysis`. Avoid a full fresh start to save prior baseline work.

Exam-Critical Facts Summary

Task 1.3: Subagent Invocation

- > Context is empty by default.
- > Pass findings as structured objects.
- > Require `source_url`, `retrieved_at`, `confidence`.
- > Preserve claim-source mapping arrays.
- > Unresolved conflicts strictly escalate to Coordinator.

Task 1.7: Session State

- > `--resume` requires explicit manual context updates.
- > `fork_session()` is for parallel exploration only.
- > Stale tool results mandate a Fresh Session.
- > Injected prior findings must be framed as hypotheses.